

Flow Models

Nagesh

May 2026

”Oh ! Dont get deceived by societal narratives of good and bad , there are no such things. Happiness and Suffering are mental events. There is No difference between doing drugs and Meditation . Beware of your biases. At the end , world is a mental representation !!!!!”
(Nagesh)

Contents

Probability Measure View of Generative Modeling	2
Normalizing Flows	2
Invertible Mapping	2
Change of Variables Theorem	2
Likelihood Learning	3
Composed Flow	3
Continuous Normalizing Flows (CNF)	3
Instantaneous Change of Variables	3
Likelihood in CNF	3
Optimal Transport View	4
Rectified Flow Models	4
Coupling Construction	4
True Velocity Field	4
Learning Objective	5
ODE Sampling	5
Unified Perspective	5
Key Insight	5
Conclusion	5

Probability Measure View of Generative Modeling

Let $(\mathcal{X}, \mathcal{B})$ be a measurable space.

We assume:

- Data distribution: $p_{\text{data}}(x)$ on $\mathcal{X}$
- Base distribution: $p_Z(z)$ on $\mathcal{Z}$ (usually $\mathbb{R}^d$)

A generative model defines a mapping:

$$f_\theta : \mathcal{Z} \rightarrow \mathcal{X}$$

which induces a **pushforward measure**:

$$p_\theta = (f_\theta)_\# p_Z$$

Meaning:

$$p_\theta(A) = p_Z(f_\theta^{-1}(A)), \quad \forall A \in \mathcal{B}$$

This is the foundation of all flow-based models.

Normalizing Flows

Invertible Mapping

Assume f_θ is a diffeomorphism:

$$x = f_\theta(z), \quad z = f_\theta^{-1}(x)$$

Change of Variables Theorem

Let $J_{f^{-1}}(x)$ be the Jacobian of inverse map.

$$p_X(x) = p_Z(z) |\det J_{f^{-1}}(x)|$$

Equivalently:

$$\log p_X(x) = \log p_Z(z) - \log |\det J_f(z)|$$

where $z = f^{-1}(x)$.

Likelihood Learning

We maximize:

$$\mathcal{L}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} [\log p_{\theta}(x)]$$

Expanding:

$$\log p_{\theta}(x) = \log p_Z(f_{\theta}^{-1}(x)) + \log \left| \det J_{f_{\theta}^{-1}}(x) \right|$$

Composed Flow

Let:

$$z_0 \sim p_Z, \quad z_K = f_K \circ \dots \circ f_1(z_0)$$

Then:

$$\log p(z_K) = \log p(z_0) - \sum_{k=1}^K \log |\det J_{f_k}(z_{k-1})|$$

Continuous Normalizing Flows (CNF)

Instead of discrete maps, define:

$$\frac{dx}{dt} = v_{\theta}(x, t)$$

with initial condition:

$$x(0) \sim p_0$$

Instantaneous Change of Variables

Let $p_t(x)$ be density at time t .

Then:

$$\frac{d}{dt} \log p_t(x(t)) = -\nabla \cdot v_{\theta}(x(t), t)$$

This comes from the continuity equation:

$$\frac{\partial p_t}{\partial t} + \nabla \cdot (p_t v_{\theta}) = 0$$

Likelihood in CNF

$$\log p_1(x(1)) = \log p_0(x(0)) - \int_0^1 \nabla \cdot v_{\theta}(x(t), t) dt$$

Optimal Transport View

We want a map transporting:

$$p_0 \rightarrow p_1$$

The Monge formulation:

$$\min_T \mathbb{E}_{x \sim p_0} \|x - T(x)\|^2 \quad \text{s.t. } T_{\#} p_0 = p_1$$

The dynamic formulation (Benamou–Brenier):

$$\min_{p_t, v_t} \int_0^1 \int \|v_t(x)\|^2 p_t(x) dx dt$$

subject to:

$$\frac{\partial p_t}{\partial t} + \nabla \cdot (p_t v_t) = 0$$

Rectified Flow Models

Coupling Construction

Sample joint pairs:

$$(x_0, x_1) \sim \pi(x_0, x_1)$$

where:

- $x_0 \sim p_0$ (noise)
- $x_1 \sim p_1$ (data)

Define interpolation path:

$$x_t = (1 - t)x_0 + tx_1$$

True Velocity Field

Differentiate:

$$\frac{dx_t}{dt} = x_1 - x_0$$

Define:

$$v^*(x, t) = \mathbb{E}[x_1 - x_0 \mid x_t = x]$$

This is the **optimal regression target**.

Learning Objective

$$\mathcal{L}(\theta) = \mathbb{E}_{(x_0, x_1) \sim \pi, t \sim U[0,1]} \left[\|v_\theta(x_t, t) - (x_1 - x_0)\|^2 \right]$$

ODE Sampling

After training:

$$\frac{dx}{dt} = v_\theta(x, t), \quad x(0) \sim p_0$$

Solve ODE to obtain:

$$x(1) \sim p_1$$

Unified Perspective

All models solve transport of probability:

$$p_0 \rightarrow p_1$$

Method	Mechanism
Normalizing Flow	Finite invertible map f_θ
CNF	Continuous velocity field + divergence tracking
Rectified Flow	Regression of transport velocity from coupling
Optimal Transport	Energy-minimizing flow paths

Key Insight

All methods share one structure:

Learn a velocity field $v(x, t)$ such that $\frac{\partial p}{\partial t} + \nabla \cdot (pv) = 0$

The differences are:

- How $v(x, t)$ is parameterized
- Whether likelihood is computed explicitly
- Whether paths are deterministic or stochastic

Conclusion

Normalizing flows, continuous flows, and rectified flows are all special cases of probability mass transport. Rectified flow can be seen as a simplified, regression-based approximation of optimal transport dynamics, while CNFs provide exact likelihood evolution through divergence tracking.